BUSINESS EMPLOYMENT TREND ANALYSIS

(2011 – 2025)

A Python-Based Data Analytics Project Featuring Statistical Analysis, Exploratory Data Analysis, and Business Visualization

A Data Analytics Capstone Project

Submitted by

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Submitted in Fulfilment of the Requirements for the

AI-Driven Data Analytics Program

Tools & Technologies

- **Python**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Seaborn**
- **Plotly**
- **Google Colab**

Dataset Source

Statistics New Zealand (Stats NZ)

Business Employment Indicators Dataset

Institution Name

Entri Elevate

Month & Year

September 2026

Abstract

Employment trends provide valuable insights into the economic performance and workforce dynamics of a country. This project analyzes New Zealand's Business Employment Indicators dataset published by Statistics New Zealand (Stats NZ) to understand employment patterns across industries between **2011 and 2025**.

The analysis focuses on identifying long-term employment growth, comparing workforce participation across industries, evaluating quarterly employment patterns, and examining the impact of Actual, Seasonally Adjusted, and Trend employment estimates. The dataset was cleaned, transformed, and analyzed using Python libraries including Pandas, NumPy, Matplotlib, Seaborn, and Plotly.

Descriptive statistics and exploratory data analysis were performed to summarize employment distributions and identify meaningful business insights. Interactive and static visualizations were developed to support objective-based analysis and highlight employment trends across industries and time periods.

The findings indicate sustained employment growth over the study period, with the Health Care and Social Assistance industry consistently recording the highest employment levels. Seasonal adjustment showed only minor differences from Actual employment estimates, indicating stable long-term workforce patterns. The results provide meaningful insights that can support workforce planning, business decision-making, and future employment policy analysis.

Keywords: Business Employment, Workforce Analysis, Employment Trends, Data Analytics, Python, Statistics New Zealand, Exploratory Data Analysis.

2. PROJECT OVERVIEW

2.1 Introduction

Employment is a key indicator of a country's economic performance and labour market conditions. Understanding employment trends across industries enables governments, businesses, and policymakers to assess workforce demand, identify growth sectors, and make informed strategic decisions. Analysing employment data over time also provides valuable insights into changes in industry performance, seasonal employment patterns, and long-term workforce development.

This project presents a comprehensive analysis of New Zealand's Business Employment Indicators dataset published by **Statistics New Zealand (Stats NZ)**. The study examines employment trends from **2011 to 2025**, focusing on filled jobs across various industries. Using Python-based data analytics techniques, the project applies statistical analysis, data preprocessing, exploratory data analysis (EDA), and interactive visualizations to identify meaningful employment patterns and business insights.

The analysis investigates employment growth over time, compares workforce participation among industries, evaluates quarterly employment variations, and examines the differences between Actual, Seasonally Adjusted, and Trend employment estimates. The findings support evidence-based decision-making by providing a clear understanding of employment dynamics across industries.

2.2 Problem Statement

Employment patterns vary across industries due to economic conditions, seasonal influences, and structural changes in the labour market. Identifying these patterns is essential for workforce planning, resource allocation, and policy formulation. However, analysing employment data over a long period presents challenges because of its large volume, multiple employment measures, and time-based variations.

This project addresses these challenges by analysing the Business Employment Indicators dataset to identify long-term employment trends, compare employment levels across industries, evaluate quarterly employment patterns, and assess the impact of different adjustment methods. The analysis aims to transform raw employment data into meaningful business insights that support strategic workforce and economic planning.

2.3 Business Objectives

The project is designed to achieve the following objectives:

Objective 1: Analyze quarterly business employment trends across industries from **2011 to 2025** to understand how employment levels have changed over time.

Objective 2: Compare employment levels across different industries to identify sectors with the highest and lowest workforce participation.

Objective 3: Examine long-term employment growth, decline, and stability across industries to identify sectors experiencing sustained growth or changing employment patterns.

Objective 4: Identify seasonal and quarterly employment patterns using **Actual**, **Seasonally Adjusted**, and **Trend** employment estimates to evaluate the effect of seasonal fluctuations on workforce levels.

2.4 Scope of the Project

The scope of this project is limited to the analysis of the Business Employment Indicators dataset published by Statistics New Zealand. The study focuses on employment information recorded between **2011 and 2025**, covering multiple industries and employment measures.

The project includes data cleaning, transformation, statistical analysis, exploratory data analysis, and visualization using Python. The analysis primarily focuses on filled jobs while excluding confidential and suppressed records to ensure data quality and reliability.

The outcomes of this study provide insights into employment growth, industry-wise workforce distribution, and seasonal employment behaviour. However, the project does not include predictive modelling or external economic indicators such as GDP, inflation, or unemployment rates.

2.5 Dataset Description

The dataset used in this project is the **Business Employment Indicators** dataset obtained from **Statistics New Zealand (Stats NZ)**. It provides quarterly information on employment across industries, regions, age groups, and other business classifications.

The dataset includes employment measures such as **Filled Jobs**, **Total Earnings**, and **Filled Jobs (Workplace Location Based)**. For this analysis, the primary focus is on **Filled Jobs**, as it directly represents employment levels across industries.

After data cleaning and preprocessing, the dataset contains variables including:

Column	Description
Series Reference	Unique identifier for each series
Period	Reporting period in quarterly format
Data Value	Employment value
Status	Final, Revised, or Confidential records

Column	Description
Group	Employment category classification
Employment Variable	Type of employment measure
Category	Industry, Region, Age Group, or Sex
Adjustment Type	Actual, Seasonally Adjusted, or Trend
Year	Extracted from the reporting period
Quarter	Quarter extracted from the reporting period

2.6 Technology Stack

The project was implemented using the following tools and technologies:

Tool / Technology	Purpose
Python	Data analysis and visualization
Google Colab	Development environment
Pandas	Data cleaning and manipulation
NumPy	Numerical computations
Matplotlib	Statistical visualization
Seaborn	Exploratory data visualization
Plotly	Interactive business dashboards and charts
Microsoft Excel	Initial data inspection

DATA PREPARATION AND PREPROCESSING

3.1 Introduction

Data preparation is a critical step in the data analytics process, as the quality of the analysis depends on the quality of the dataset. Raw datasets often contain missing values, inconsistent formatting, duplicate records, and unnecessary variables that can affect the accuracy of analytical results. Therefore, the dataset was carefully examined, cleaned, and transformed before performing statistical analysis and visualization.

The Business Employment Indicators dataset obtained from Statistics New Zealand (Stats NZ) was preprocessed using Python to improve data quality, ensure consistency, and prepare it for meaningful analysis.

3.2 Data Collection

The dataset used in this project was obtained from **Statistics New Zealand (Stats NZ)**, which publishes official quarterly Business Employment Indicators. The dataset contains employment information across multiple industries, regions, demographic groups, and employment measures.

The data spans from **2011 to 2026**, with quarterly observations representing workforce statistics across different employment categories. For this project, the analysis focuses on employment records from **2011 to 2025**, as the 2026 data was incomplete.

3.3 Initial Dataset Overview

Before preprocessing, the dataset was examined to understand its structure and identify potential quality issues.

The initial inspection included:

- Number of rows and columns
- Data types
- Missing values
- Duplicate records
- Unique values
- Statistical summary

3.4 Data Cleaning

The following data cleaning activities were performed to improve the quality and reliability of the dataset.

3.4.1 Handling Missing and Suppressed Values

The dataset contained suppressed records where employment values were unavailable for confidentiality reasons.

Instead of imputing these values, the suppressed records were removed because the actual employment values were unknown. Imputing such values could introduce bias and reduce the reliability of the analysis.

3.4.2 Removing Unnecessary Columns

Columns that did not contribute to the analysis were removed to simplify the dataset.

The following columns were dropped:

- Subject
- Series_title_4
- Series_title_5

Removing these variables reduced redundancy and improved the overall structure of the dataset.

3.4.3 Renaming Columns

Several column names were renamed to improve readability and simplify analysis.

Original Column	Renamed Column
Series_title_1	Employment_Variable
Series_title_2	Category
Series_title_3	Adjustment_Type

These descriptive names made the dataset easier to understand and use throughout the project.

3.4.4 Feature Engineering

Additional variables were created from the existing **Period** column.

The following new features were generated:

- **Year**
- **Quarter**

These variables enabled yearly and quarterly trend analysis throughout the project.

3.4.5 Data Type Validation

The dataset was checked to ensure that all variables had appropriate data types.

Examples:

- Year → Integer
- Data Value → Float
- Category → Object
- Adjustment Type → Object

Validating data types ensured accurate grouping, aggregation, and visualization.

3.5 Final Cleaned Dataset

After preprocessing, the cleaned dataset (**df_clean**) was used for all statistical analyses and visualizations.

The cleaned dataset contained:

- Clean employment records
- Consistent column names
- Derived Year and Quarter variables
- No suppressed employment values
- Improved data quality for analysis

This prepared dataset served as the foundation for all subsequent analyses.

4. STATISTICAL ANALYSIS

4.1 Introduction

Statistical analysis provides a numerical summary of the dataset and helps understand the distribution, central tendency, and variability of employment values before performing exploratory data analysis. It enables meaningful interpretation of workforce patterns and supports evidence-based business decisions.

In this chapter, descriptive statistical techniques are applied to the **Data_value** variable of the Business Employment Indicators dataset. The analysis includes measures of central tendency and measures of dispersion, supported by appropriate visualizations to illustrate the distribution and spread of employment values.

4.2 Measure of Central Tendency

Measures of central tendency describe the central or typical value of a dataset. They provide an overall understanding of where the majority of observations are concentrated. In this project, the **mean**, **median**, and **mode** were calculated to summarize the distribution of employment values. A histogram was used to visualize the distribution and compare these statistical measures.

Figure 4.1: Distribution of Employment Values with Mean, Median, and Mode

Analysis Objective

To examine the overall distribution of employment values and identify the central tendency of the dataset using the mean, median, and mode.

Variables Used

Component	Description
Dataset	Business Employment Indicators (Stats NZ)
Variable Analyzed	Data_value
Chart Type	Histogram with Mean, Median, and Mode
Statistical Measures	Mean, Median, Mode

Chart Interpretation

- The mean employment value (74,018) is significantly higher than the median (14,265) and mode (287), indicating that a few extremely large employment values are increasing the overall average.
- The pattern Mean > Median > Mode confirms that the employment data is positively (right) skewed, showing that employment is not evenly distributed across industries and categories.

- The median provides a more reliable measure of a typical employment value because it is less affected by extreme employment figures than the mean.
- The mode suggests that lower employment values occur most frequently, indicating that many industries or categories have relatively smaller workforces compared to a few dominant sectors.
- From a business perspective, employment is concentrated in a limited number of high-employment industries, while the majority of industries operate with comparatively smaller workforce sizes. Therefore, industry-level analysis is essential for making accurate workforce and policy decisions.

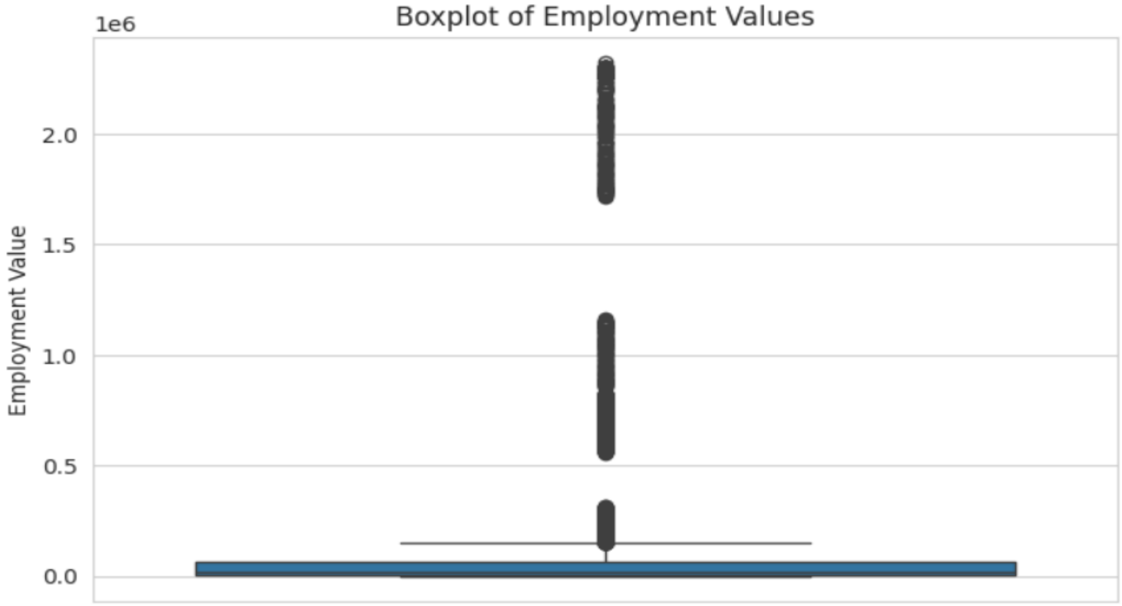
Key Findings

- Employment values are not evenly distributed across the dataset.
- A small number of industries account for very high employment values.
- The median provides a better representation of the typical employment level than the mean.
- The histogram confirms that employment data is concentrated at lower values with a smaller number of high-value observations.

4.3 Measure of Dispersion

Measures of dispersion describe the variability or spread of data around the central value. They help determine whether employment values are closely clustered or widely dispersed. In this project, **variance** and **standard deviation** were calculated, while a box plot was used to visualize the spread of employment values and identify potential outliers.

Figure 4.2: Box Plot of Employment Values



Analysis Objective

To evaluate the variability of employment values and identify the presence of outliers using statistical measures of dispersion.

Chart Interpretation

- The box plot reveals a wide spread in employment values, indicating substantial variation in workforce sizes across different industries and business categories.
- The wide interquartile range (IQR) shows that the middle 50% of employment values are spread over a large range, confirming that employment levels are not uniformly distributed.
- A large number of observations appear above the upper whisker as outliers, showing that a few industries have exceptionally high employment levels compared to the majority of the dataset.
- The median is positioned closer to the lower quartile, while the upper whisker extends much farther than the lower whisker, confirming that the data is positively skewed.
- These outliers represent genuine business observations rather than data errors, as they correspond to large industries with naturally higher employment levels. Therefore, they should be retained for meaningful business analysis instead of being removed.
- From a business perspective, the box plot indicates that employment is concentrated in a limited number of high-employment sectors, while most industries maintain comparatively smaller workforces. This uneven distribution highlights the importance of analyzing employment trends by industry rather than relying solely on overall summary statistics.

Key Findings

- Employment values exhibit high variability across industries.
- Several industries record exceptionally high employment compared to the rest of the dataset.
- The box plot confirms the presence of outliers.
- Measures of dispersion indicate that workforce distribution is highly uneven.

5.EXPLORATORY DATA ANALYSIS (EDA)

5.1 Introduction

- Exploratory Data Analysis (EDA) is a fundamental step in data analytics that involves examining, summarizing, and visualizing data to identify meaningful patterns, trends, relationships, and anomalies. It transforms raw numerical information into actionable business insights through statistical summaries and graphical representations.
- In this project, EDA was conducted on the Business Employment Indicators dataset to understand employment patterns across industries in New Zealand from **2011 to 2025**.

Various visualization techniques were used to analyse employment trends, compare workforce participation among industries, examine long-term employment growth, and evaluate seasonal employment patterns. The analysis directly addresses the project objectives and supports informed business decision-making.

5.2 Analysis Framework

The exploratory analysis is organized according to the four business objectives defined for this project.

Section	Project Objective	Analysis Type
5.2	Objective 1: Analyze quarterly business employment trends across industries (2011–2025).	Univariate & Bivariate
5.3	Objective 2: Compare employment levels across industries.	Bivariate
5.4	Objective 3: Examine long-term employment growth across industries.	Bivariate & Multivariate
5.5	Objective 4: Identify seasonal and quarterly employment patterns.	Multivariate

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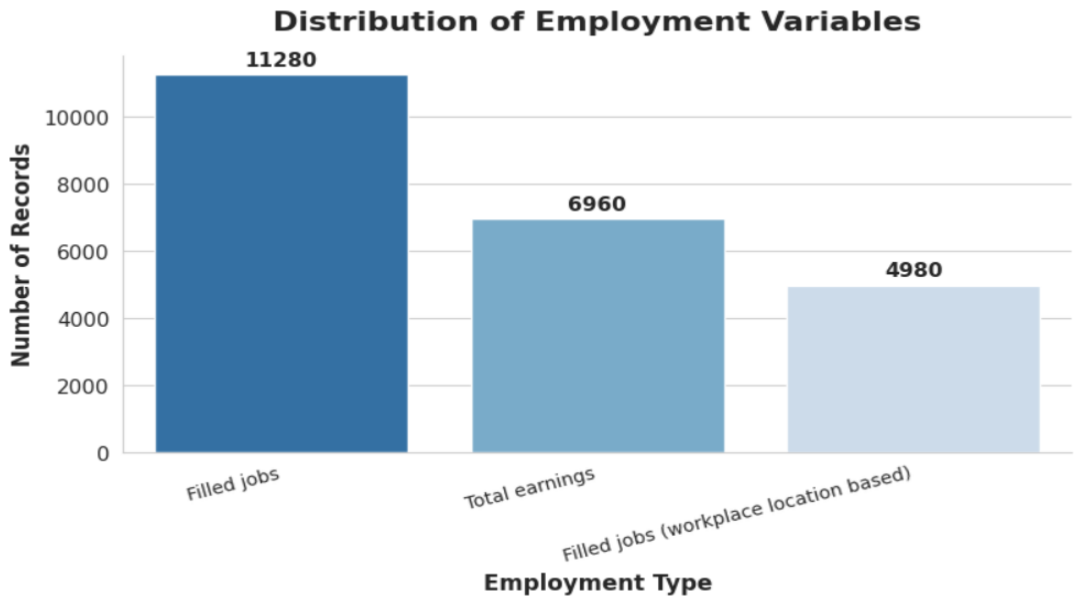
5.2 Objective 1: Analyze Quarterly Business Employment Trends Across Industries (2011–2025)

Objective Statement

To examine how business employment levels have changed over time by analysing yearly and quarterly employment trends from 2011 to 2025.

Visualizations

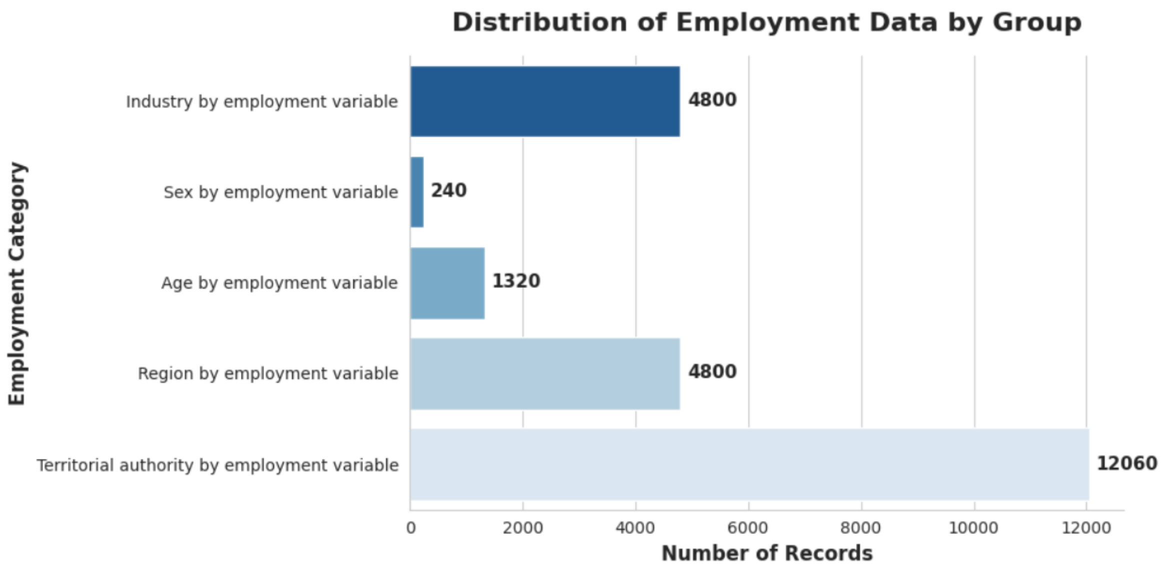
5.2.1 Distribution of Employment Variables



Key Inights:

- **Filled Jobs** is the most frequently recorded employment variable, with **11,280 records**, accounting for nearly half of the dataset.
- **Total Earnings** contains **6,960 records**, indicating that earnings-related information is available for a substantial portion of the observations.
- **Filled Jobs (Workplace Location Based)** has the lowest frequency, with **4,980 records**, suggesting that workplace-based employment data is less extensively represented.
- The unequal distribution of employment variables indicates that the dataset places greater emphasis on **employment counts** than on earnings or workplace-based employment measures.
- The high availability of **Filled Jobs** records supports its selection as the primary variable for subsequent employment trend and industry-level analyses.

5.2.2 Distribution of Groups



Key Insights

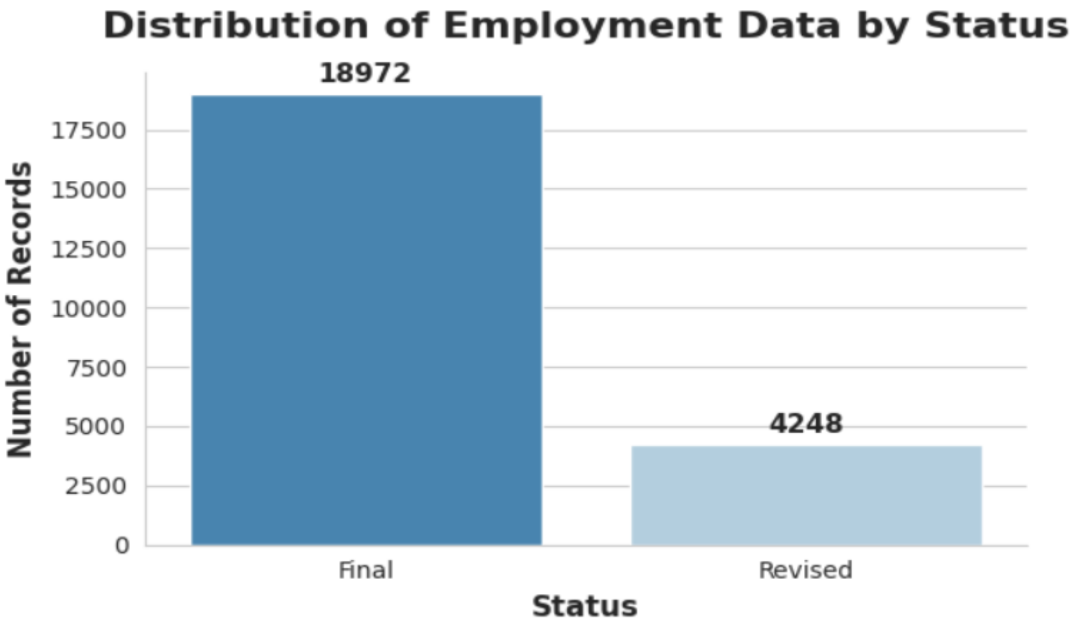
- **Territorial Authority by Employment Variable** contains the highest number of records (**12,060**), indicating that the dataset provides the most detailed employment information at the local government level.

- **Industry by Employment Variable** and **Region by Employment Variable** each contain **4,800 records**, demonstrating substantial coverage for both industry-specific and regional employment analyses.

- **Age by Employment Variable** comprises **1,320 records**, providing moderate demographic information for workforce analysis across different age groups.

- **Sex by Employment Variable** has the fewest records (**240**), indicating that gender-based employment information represents a relatively small portion of the dataset.
- The distribution of records shows that the dataset primarily emphasizes **geographical (territorial authority and region)** and **industry-level employment data**, making these groups the most suitable for detailed employment trend and workforce analyses.

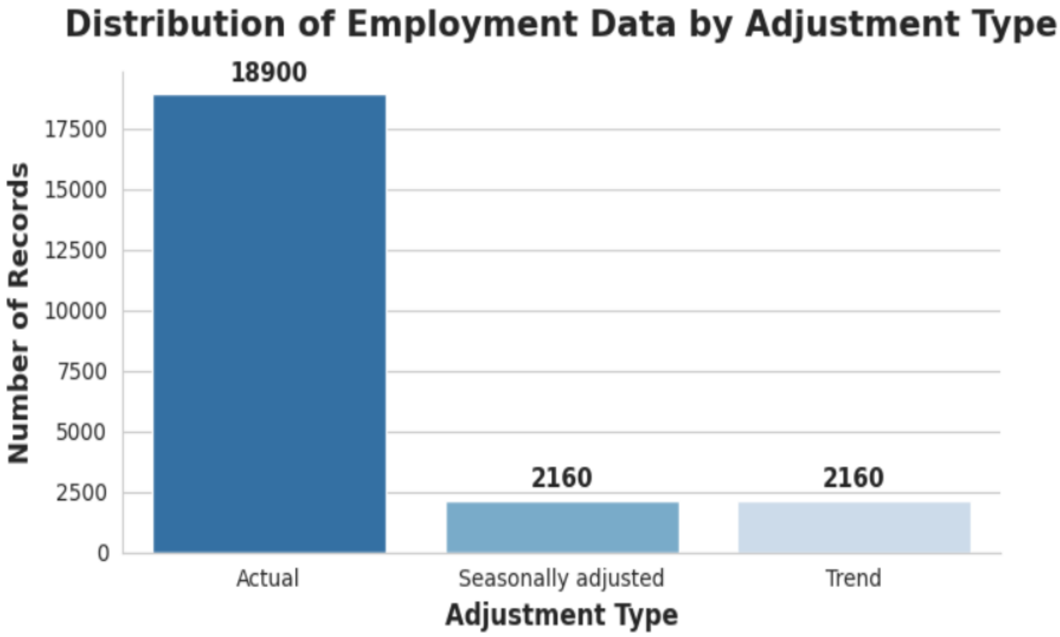
5.2.3 Distribution of Status



Key Insights

- **Final** records dominate the dataset with **18,972 observations**, accounting for the majority of employment data available for analysis.
- **Revised** records total **4,248 observations**, indicating that a smaller proportion of employment estimates were updated after their initial release.
- The significant difference between Final and Revised records suggests that most employment estimates were finalized, providing a stable and reliable dataset for analysis.
- The relatively limited number of Revised records indicates that only a small portion of employment data required subsequent corrections or updates.
- The predominance of Final records enhances the credibility of the dataset and supports the reliability of the employment trends and industry comparisons presented in this study.

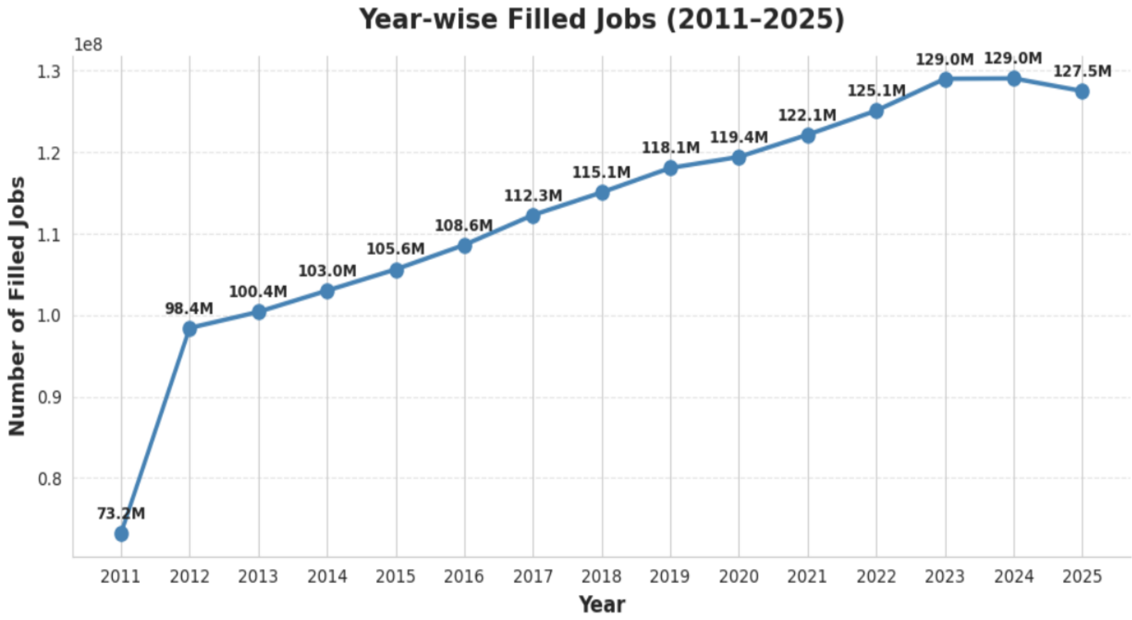
5.2.4 Distribution of Employment Data by Adjustment Type



Key Insights

- **Actual** employment data constitutes the largest portion of the dataset with **18,900 records**, indicating that most analyses are based on observed employment values.
- **Seasonally Adjusted** and **Trend** data each contain **2,160 records**, providing an equal representation for analyzing seasonal effects and long-term employment patterns.
- The predominance of **Actual** records suggests that the dataset primarily focuses on reporting real employment outcomes rather than adjusted estimates.
- The availability of **Seasonally Adjusted** and **Trend** data enables more accurate assessment of employment patterns by minimizing seasonal fluctuations and highlighting long-term trends.
- The balanced representation of all three adjustment types provides flexibility to perform different forms of employment analysis, including actual performance, seasonal comparison, and long-term trend evaluation.

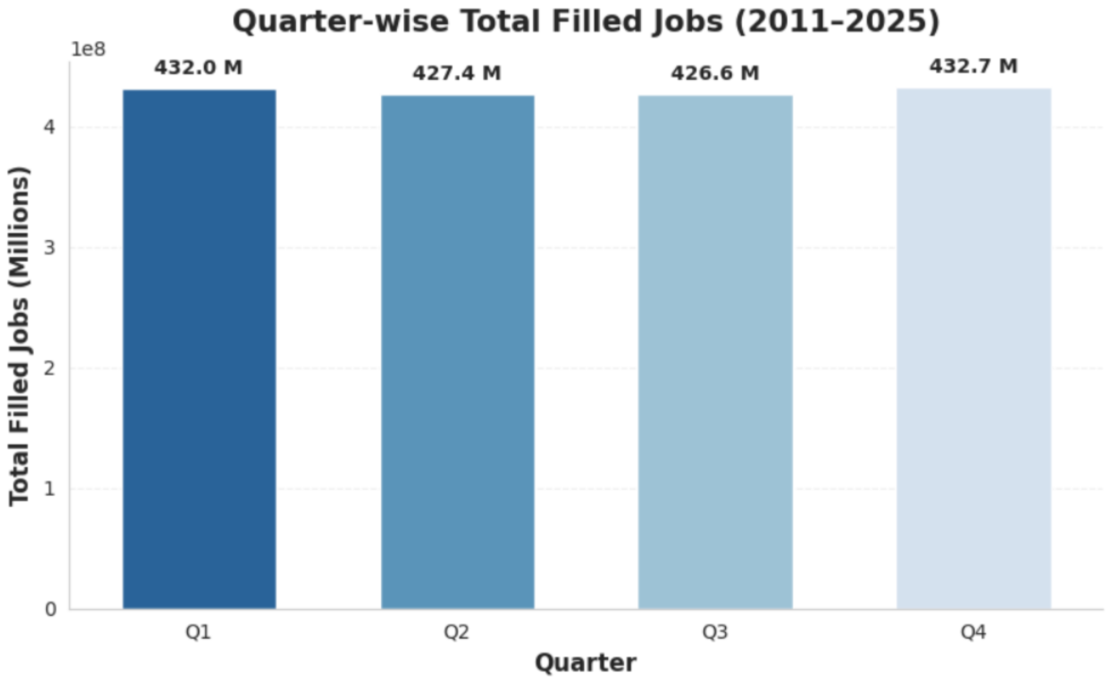
5.2.5 Year-wise Employment Trend



Key Insights

- **Filled jobs exhibited a strong long-term upward trend**, increasing from **73.2 million in 2011** to a peak of **129.0 million in both 2023 and 2024**, reflecting sustained employment growth over the study period.
- **Employment grew consistently between 2012 and 2022**, with year-on-year increases indicating a stable expansion of workforce participation across industries.
- **The highest employment levels were recorded in 2023 and 2024 (129.0 million)**, suggesting that these years represented the strongest employment performance during the analysis period.
- **A slight decline to 127.5 million was observed in 2025**, representing a marginal decrease from the previous two years while remaining significantly higher than the employment levels recorded in earlier years.
- **Overall, the analysis indicates a positive long-term employment trajectory**, demonstrating continuous workforce growth across New Zealand businesses from 2011 to 2025, with only a minor moderation in the final year.

5.2.5 Quarter-wise Filled Jobs



Key Insights

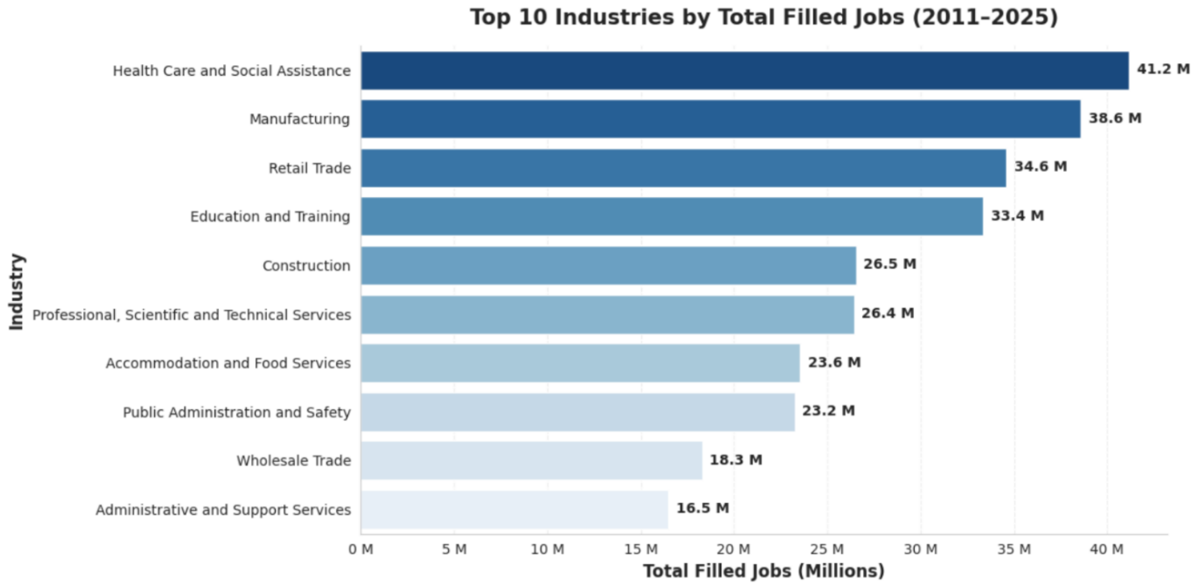
- **Quarter 4 (Q4)** recorded the highest total filled jobs at **432.7 million**, indicating the strongest employment activity across the study period.
- **Quarter 3 (Q3)** had the lowest total filled jobs at **426.6 million**, although the difference from the other quarters was relatively small.
- The variation between the highest and lowest quarterly employment totals is only **6.1 million jobs**, suggesting that employment remained **stable throughout the year** with limited seasonal fluctuations.
- **Quarter 1 (432.0 million)** and **Quarter 4 (432.7 million)** exhibited slightly higher employment levels than **Quarter 2 (427.4 million)** and **Quarter 3 (426.6 million)**, indicating marginal increases at the beginning and end of the year.
- Overall, the analysis suggests that **business employment remained consistently high across all four quarters from 2011 to 2025**, reflecting a stable labour market with only minor seasonal variations.

5.3 Objective 2: Compare Employment Levels Across Industries

Objective Statement

To identify industries with the highest workforce participation by comparing total filled jobs across industry sectors.

5.3.1 Industry-wise Employment Distribution



Key Insights

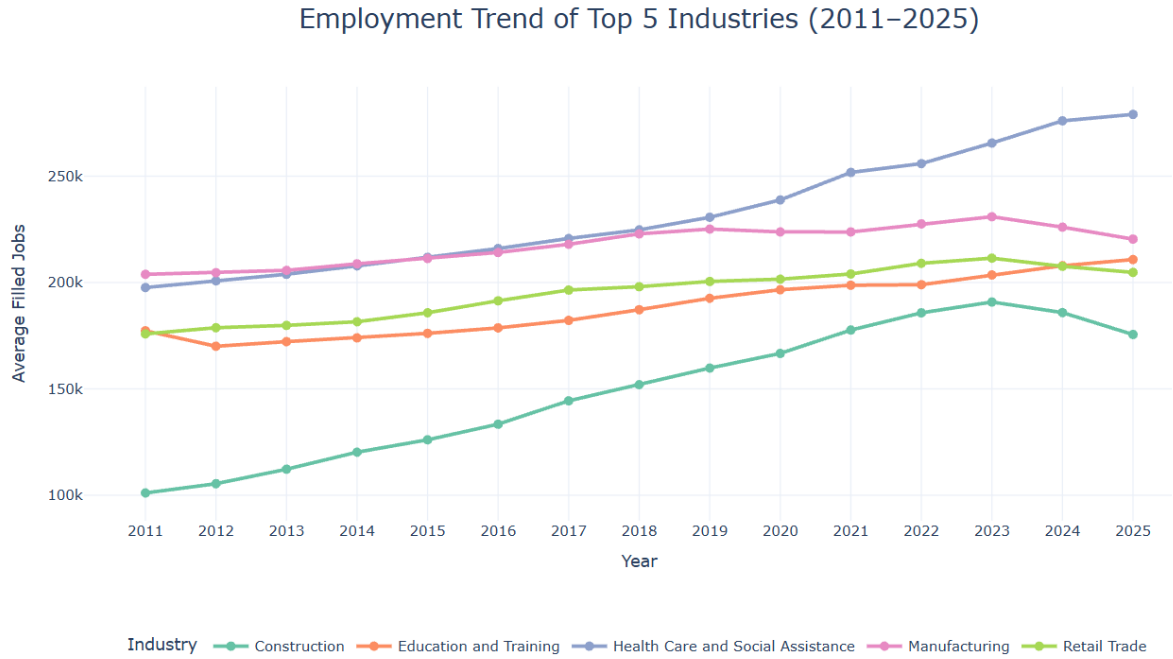
- **Health Care and Social Assistance** recorded the highest employment, with **41.2 million filled jobs**, making it the largest employer among the top ten industries during 2011–2025.
- **Manufacturing (38.6 million)** and **Retail Trade (34.6 million)** ranked second and third, highlighting their significant contribution to overall employment.
- **Education and Training (33.4 million)** also maintained a strong workforce, reflecting the continued demand for education services throughout the study period.
- The lower-ranked industries within the top ten, including **Wholesale Trade (18.3 million)** and **Administrative and Support Services (16.5 million)**, employed considerably fewer workers than the leading sectors, indicating substantial variation in workforce size across industries.
- Overall, the analysis demonstrates that employment is **concentrated in service-oriented industries**, particularly **Health Care and Social Assistance, Retail Trade, and Education and Training**, while other industries contribute comparatively smaller shares to total employment.

5.4 Objective 3: Examine Long-Term Employment Growth Across Industries

Objective Statement

To evaluate long-term employment growth, decline, and stability across industries between 2011 and 2025.

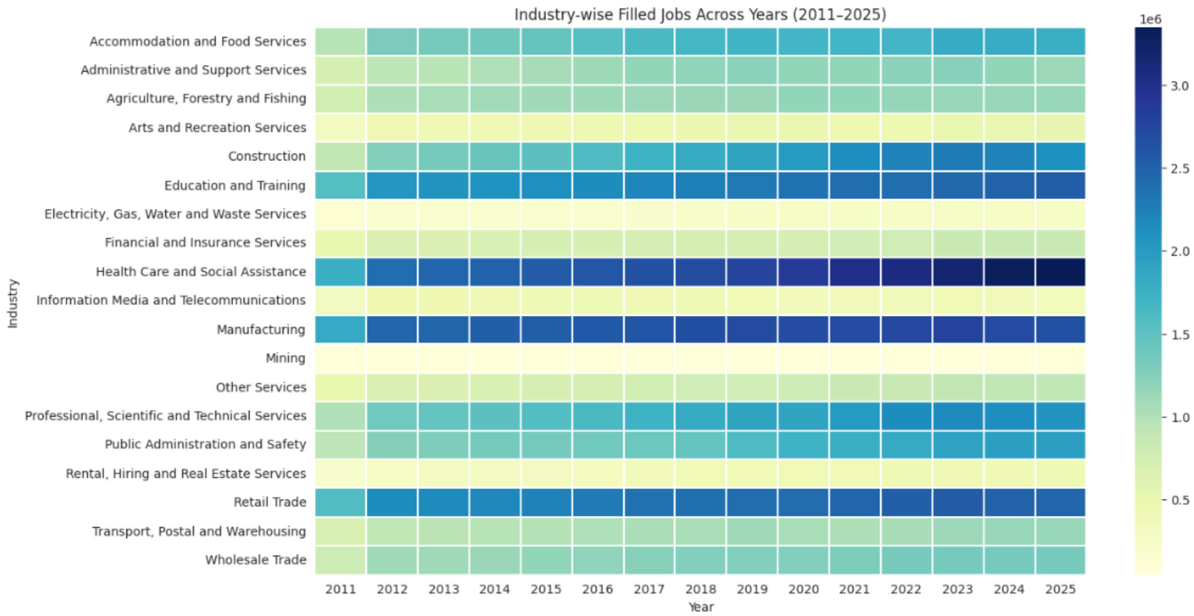
5.4.1 Employment Trend of Top 5 Industries



Key Insights

- **Health Care and Social Assistance** experienced the **strongest and most consistent employment growth**, increasing from approximately **198,000** average filled jobs in 2011 to **278,000** in 2025, making it the fastest-growing industry over the study period.
- **Construction** recorded the **highest percentage growth** among the five industries, rising steadily from around **101,000** jobs in 2011 to **175,000** jobs in 2025, reflecting sustained expansion in the construction sector despite a slight decline in the final year.
- **Education and Training** demonstrated **stable and continuous employment growth**, increasing from approximately **176,000** to **211,000** average filled jobs, indicating consistent demand for educational services.
- **Manufacturing** maintained a relatively **stable workforce** throughout the period, with gradual growth until **2023**, followed by a moderate decline during **2024–2025**, suggesting a slowdown after years of steady employment.
- **Retail Trade** showed **moderate long-term growth**, increasing steadily until **2023**, before experiencing a slight decline in **2024–2025**, indicating that employment remained resilient while moderating in the later years.

5.4.2 Examine employment growth across all industries over multiple years.



Key Insights

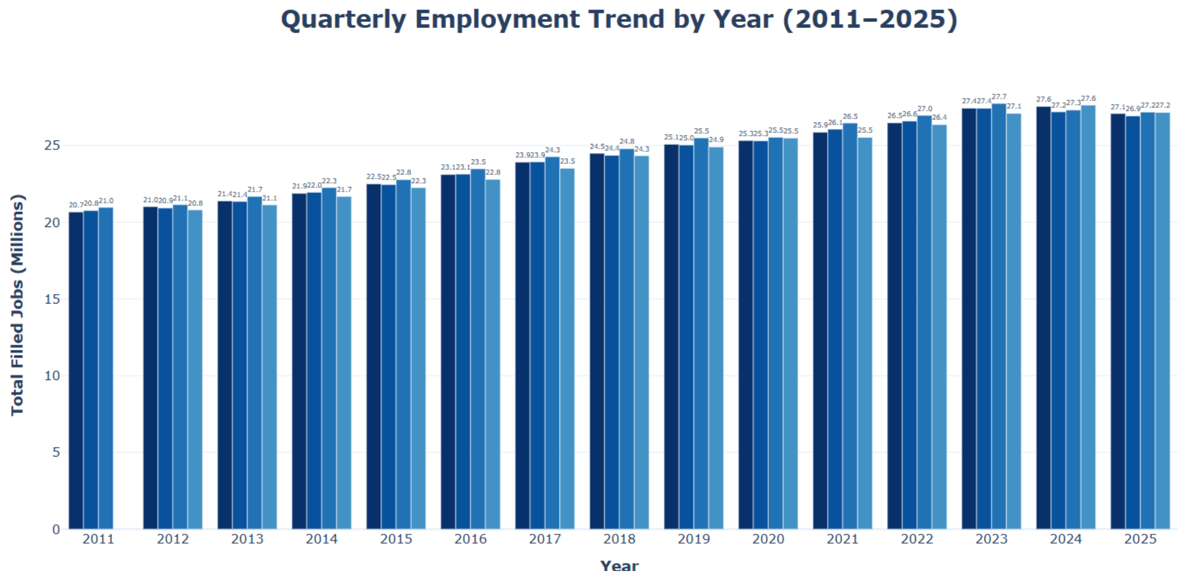
- **Health Care and Social Assistance** consistently recorded the **highest employment levels** throughout 2011–2025, with color intensity becoming progressively darker over time, indicating sustained workforce expansion.
- **Manufacturing, Retail Trade, and Education and Training** maintained **consistently high employment** across all years, demonstrating stable workforce participation despite minor year-to-year fluctuations.
- **Construction** showed a **clear upward employment trend**, with darker shades appearing after 2016, reflecting strong long-term growth driven by increased infrastructure and development activities.
- **Mining** and **Electricity, Gas, Water and Waste Services** remained among the **lowest employing industries**, as indicated by consistently lighter color shades throughout the study period, suggesting relatively smaller workforce requirements.
- Most industries exhibit a **gradual transition from lighter to darker shades over time**, indicating that employment generally increased across sectors between 2011 and 2025 rather than being concentrated in only a few industries.

5.5 Objective 4: Identify Seasonal and Quarterly Employment Patterns

Objective Statement

To evaluate seasonal employment behaviour by comparing quarterly employment patterns and adjustment type.

5.5.1 Compare quarterly employment patterns for each year.



Key Insights – Quarterly Employment Trend by Year (2011–2025)

1. **Employment increased consistently across all quarters from 2011 to 2024**, indicating sustained business expansion and a steady rise in workforce demand throughout the study period.

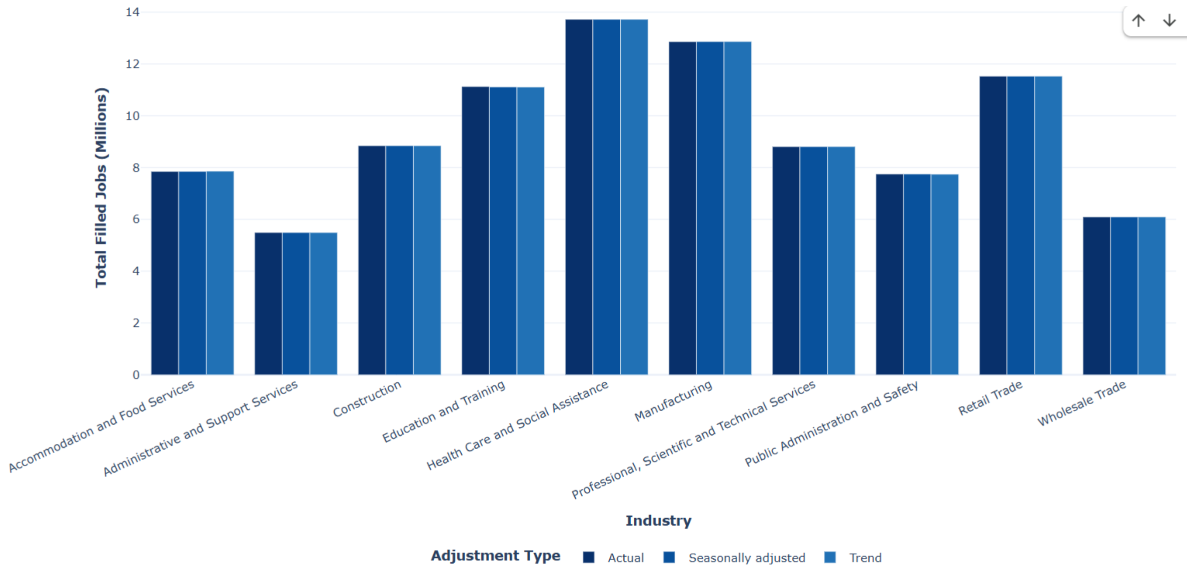
2. **Quarter 4 (Q4) generally recorded the highest employment levels in most years**, reaching approximately **27.6 million filled jobs in 2024**, suggesting increased employment activity toward the end of the year.

3. **Quarter 3 (Q3) consistently showed the lowest employment among the four quarters**, although the difference from other quarters remained relatively small, indicating only modest seasonal fluctuations.

4. **The highest quarterly employment levels were observed during 2023 and 2024**, with all four quarters exceeding **27 million filled jobs**, reflecting the strongest labour market performance during the analysis period.

5. **A slight decline is visible in 2025 across all quarters**, with employment decreasing from the 2024 peak. This suggests a temporary moderation in employment growth rather than a major downward trend, as quarterly employment remained above **27 million filled jobs**.

5.5.2 Compare Actual, Seasonally Adjusted, and Trend employment estimates across industries.



Key Insights – Industry-wise Employment by Adjustment Type (2011–2025)

- **Health Care and Social Assistance** consistently records the **highest employment levels** across all three adjustment types (Actual, Seasonally Adjusted, and Trend), highlighting its position as the largest employment-generating industry during the study period.
- **Manufacturing** ranks as the **second-largest employer**, followed by **Retail Trade** and **Education and Training**, indicating these sectors maintain a strong and stable workforce over time.
- The employment values for **Actual, Seasonally Adjusted, and Trend** are **very similar within each industry**, suggesting that seasonal fluctuations have only a **minimal impact** on long-term employment patterns.
- Industries such as **Wholesale Trade** and **Administrative and Support Services** record comparatively lower employment among the top ten industries, although they still contribute significantly to overall business employment.
- The close alignment between the three adjustment types indicates that the dataset is **highly stable**, with seasonal adjustment producing only minor variations from the actual employment figures.
- Overall, the analysis demonstrates that **industry-specific workforce size has a much greater influence on employment levels than seasonal effects**, making long-term industry performance the primary driver of employment trends.

Findings and Business Insights

6.1 Key Findings

Finding 1 – Strong Long-Term Employment Growth

Filled jobs increased consistently from **73.2 million in 2011** to **127.5 million in 2025**, representing substantial long-term employment growth. Employment reached its highest level in **2023 and 2024 (129.0 million)** before experiencing a slight decline in 2025, indicating a stable labour market with minor recent fluctuations.

Finding 2 – Limited Seasonal Variation

Quarterly employment remained relatively stable throughout the analysis period. Total employment across all quarters ranged between **426.6 million and 432.7 million**, suggesting that employment patterns are only marginally affected by seasonal changes.

Finding 3 – Health Care and Social Assistance is the Largest Employer

Among all industries, **Health Care and Social Assistance** consistently recorded the highest employment levels throughout the study period. Continuous workforce expansion reflects increasing demand for healthcare and social support services.

Finding 4 – Manufacturing Remains a Core Employment Sector

Manufacturing maintained one of the largest workforces across all years. Although employment growth slowed during the later years, the sector remained one of the country's major employment contributors.

Finding 5 – Construction Experienced Rapid Growth

Construction demonstrated one of the strongest growth trends over the fifteen-year period. Employment expanded steadily from 2011 until 2023, highlighting increased infrastructure development and construction activities.

Finding 6 – Education and Training Shows Stable Expansion

The Education and Training sector displayed continuous year-on-year employment growth with very limited fluctuations, indicating sustained investment in educational services.

Finding 7 – Retail Trade Maintained High Workforce Levels

Retail Trade consistently ranked among the top employing industries. While moderate fluctuations occurred during the final years, the sector maintained a large and stable workforce.

Finding 8 – Adjustment Methods Produce Similar Results

Comparison of **Actual**, **Seasonally Adjusted**, and **Trend** employment estimates showed only minor differences across industries. This indicates that seasonal effects have minimal influence on overall employment patterns and confirms the reliability of long-term employment trends.

Finding 9 – Data Quality Supports Reliable Analysis

The dataset consisted predominantly of **Final** status records, with relatively fewer Revised records. The cleaned dataset contained consistent data types and removed suppressed records, improving overall analytical reliability.

Finding 10 – Employment Growth Was Broad-Based

The heatmap and multivariate analyses indicate that employment growth occurred across nearly all industries rather than being concentrated in only a few sectors. Service industries generally experienced stronger long-term expansion compared to several traditional industries.

6.2 Overall Business Insight

The Business Employment dataset demonstrates that New Zealand's labour market experienced sustained employment growth between 2011 and 2025. Healthcare, Manufacturing, Retail Trade, Education, and Construction emerged as the principal employment-generating sectors. Quarterly employment remained relatively stable, indicating limited seasonal variation, while adjustment methods confirmed the consistency of long-term employment trends. Overall, the analysis suggests a resilient labour market supported by continuous expansion in service-oriented industries and steady workforce growth across most sectors.

CONCLUSION AND RECOMMENDATIONS

7.1 Conclusion

This project analyzed quarterly business employment data from New Zealand covering the period **2011–2025** using Python-based data analytics techniques. The dataset was cleaned, transformed, and explored through descriptive statistics and various univariate, bivariate, and multivariate visualizations to understand long-term employment trends across industries.

The analysis revealed that overall employment experienced a consistent upward trend throughout the study period, indicating sustained workforce growth across most industries. Health Care and Social Assistance, Manufacturing, Retail Trade, and Education and Training emerged as the largest employment-generating sectors, while industries such as Mining and Electricity, Gas, Water and Waste Services maintained comparatively smaller workforce sizes.

Quarterly analysis showed only minor seasonal fluctuations, suggesting that employment remained relatively stable throughout the year. The comparison between Actual, Seasonally Adjusted, and Trend employment estimates demonstrated that seasonal effects had only a limited impact on long-term employment patterns, confirming the reliability of the observed growth trends.

Overall, the project successfully achieved all research objectives by identifying employment growth patterns, comparing industry performance, evaluating long-term trends, and examining seasonal variations using statistical and visualization techniques. The findings provide valuable insights into workforce distribution and employment dynamics across New Zealand industries.

7.2 Recommendations

Based on the findings of this study, the following recommendations are suggested:

• Strengthen Workforce Planning
Industries experiencing continuous employment growth, particularly Health Care and Social Assistance, Construction, and Education and Training, should continue workforce planning initiatives to meet future labor demand.

• Monitor Declining or Stable Industries
Industries showing slower employment growth or stagnation should be monitored regularly to identify potential operational or economic challenges affecting workforce expansion.

• Utilize Trend Data for Strategic Decisions
Trend-adjusted employment estimates should be preferred for long-term planning because they minimize seasonal fluctuations and better represent underlying employment growth.

• Enhance Quarterly Performance Monitoring
Organizations and policymakers should continue reviewing quarterly employment patterns to detect early signs of labor market changes and respond proactively.

• Support High-Growth Industries
Government agencies and business organizations should prioritize investment, skill development, and workforce training programs in industries demonstrating sustained employment growth.

• Expand Future Research
Future studies may incorporate additional business indicators such as wages, business births and closures, inflation, GDP, or unemployment rates to provide a broader understanding of employment dynamics.

7.3 Limitations of the Study

Although the analysis provides meaningful insights, several limitations should be acknowledged:

- The study is based solely on the **Stats NZ Business Employment Indicators dataset**.
- Only employment-related variables available in the dataset were analyzed.
- External economic events and policy changes were not incorporated into the analysis.
- The 2026 data was excluded because only the first quarter (Q1) was available, making year-to-year comparisons incomplete.
- The study primarily uses descriptive statistical techniques and visualization methods rather than predictive or machine learning models.

7.4 Future Scope

Future work can extend this project by:

- Developing employment forecasting models using machine learning techniques.
- Comparing employment trends across multiple countries.
- Integrating economic indicators such as GDP growth, inflation, and unemployment rates.
- Performing predictive analysis to estimate future employment demand across industries.